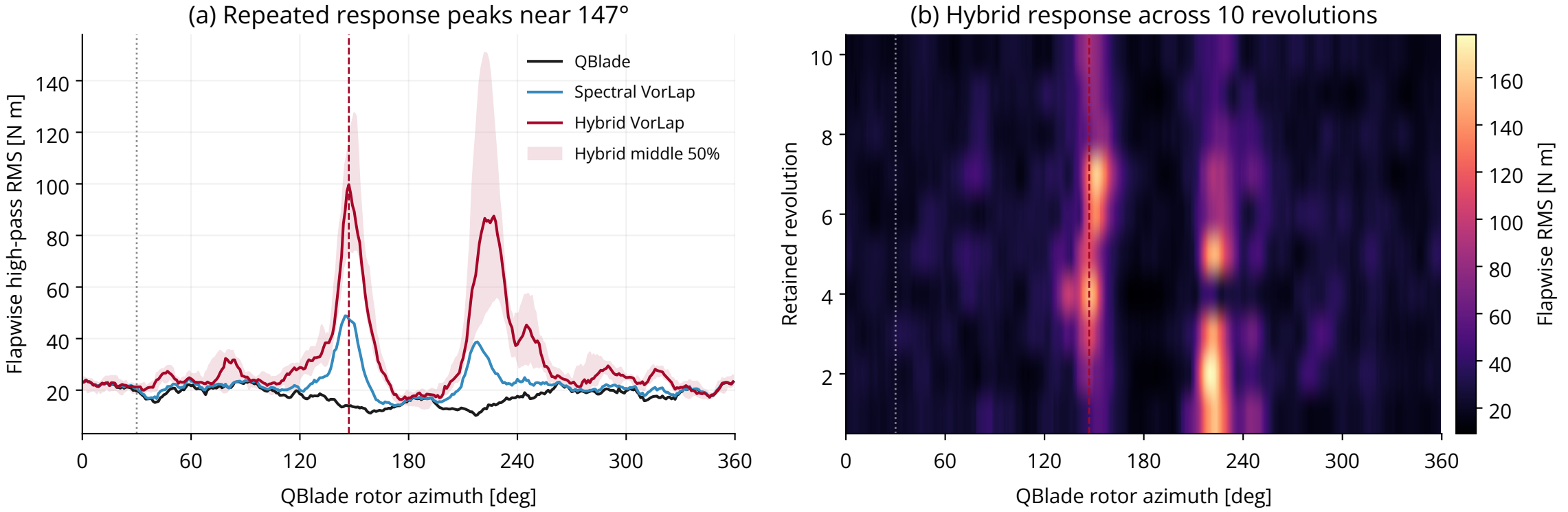

(a) Repeated response peaks near 147°
QBlade
Spectral VorLap
Hybrid VorLap
Hybrid middle 50%
Flapwise high-pass RMS [N m]
20
40
60
80
100
120
140
0
60
120
180
240
300
360
QBlade rotor azimuth [deg]
(b) Hybrid response across 10 revolutions
Retained revolution
2
4
6
8
10
0
60
120
180
240
300
360
QBlade rotor azimuth [deg]
Flapwise RMS [N m]
20
40
60
80
100
120
140
160